Xiao Jiang (江筱)

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Research Summary & Interests

I am a recent Ph.D. graduate in condensed matter physics, focusing on the fundamental physics of electron–phonon interactions and polaron quasiparticles. During my Ph.D., I used first-principles methods (DFT, DFPT, Wannier interpolation, EPW) to characterise polaron formation and transport in transition-metal oxides. I established which polaron types form and identified the microscopic mechanisms driving their formation.

While DFPT/EPW can evaluate the lowest-order (Fan–Migdal) electron–phonon self-energy, the polaron problem in strongly coupled systems may require summing Feynman diagrams beyond the Migdal approximation, including vertex corrections and multi-phonon dressing effects that are difficult to access in standard first-principles workflows. To address this limitation, I designed and implemented **`PolaronTCISolver`**, a C++23 solver based on the Quantics Tensor Cross Interpolation (QTCI) framework. The engine evaluates selected single-band electron–phonon Feynman diagrams by compressing high-dimensional integrands into Tensor Train form, providing a deterministic numerical route with controlled tolerances and cross-validation at each diagram order. Additionally, I developed **`CrystalCanvas`**, an open-source GPU-accelerated scientific visualization application for crystal structures, volumetric fields, phonons, Wannier models, and reciprocal space — built for publication-quality figures.

My research trajectory bridges established first-principles materials modelling with numerically controlled many-body diagrammatic techniques, focusing on three future directions:

- **Many-Body Diagrammatic Methods for Polaron Systems:** Extending TCI-based approaches to evaluate higher-order Feynman diagrams for polaron self-energy in realistic multi-orbital materials, moving beyond model Hamiltonians toward first-principles accuracy.
- **Electron–Phonon Coupling in Strongly Correlated Materials:** Combining first-principles EPC parameters with many-body frameworks (DMFT, diagrammatic methods) to capture the interplay between strong electron–electron correlations and lattice dynamics.
- **Computational Design of Quantum Materials via EPC:** Applying advanced electron–phonon coupling calculations to predict emergent phenomena, including conventional and unconventional superconductivity, charge-density waves, and polaron-mediated transport in candidate materials.

Academic Career and Education

07/2025–present **Independent Research and Numerical Method Development**, China.
Core Developer: `PolaronTCISolver` and `CrystalCanvas`.

09/2018–06/2025 **Doctoral Researcher**, Lanzhou University, China.
Ph.D. in Physics (Condensed Matter Physics), 2025
Dissertation: "First-principles simulation study of carrier transport in transition metal oxide photocatalytic materials"
Advisors: Prof. Weihua Han and Prof. Zemin Zhang
Initial doctoral research focused on experimental materials studies. Transitioned in 2020 to first-principles computational research, initially studying electronic structure and subsequently focusing on electron–phonon interactions and polaron formation.

09/2016–08/2018 **Graduate Student (Master's Stage, Combined MS–PhD Programme)**, Lanzhou University, China.
Research training in materials synthesis, photoelectrochemical characterisation and transmission electron microscopy.

09/2012–06/2016 **B.S. in Physics**, Lanzhou University, China.

Independent Scientific Software

`PolaronTCISolver` 07/2025–present

A C++23 many-body solver built to compute the polaron Green's function $G(\mathbf{k}, \tau)$ and self-energy $\Sigma(\mathbf{k}, i\omega_n)$ via high-order Feynman diagrams.

- Leverages Quantics Tensor Cross Interpolation (QTCI) as a deterministic numerical route for selected diagram evaluations, with controlled tolerances and cross-validation at each diagram order.
- Compresses the full momentum- and imaginary-time-dependent integrand into Tensor Train format, yielding $\Sigma(\mathbf{k}, i\omega_n)$ across the selected grid in a single deterministic solve. A prototype multi-orbital matrix-product extension is implemented.
- Includes prototype post-processing components for Discrete Lehmann Representation (DLR) basis compression, the Dyson equation, and analytic continuation, together with spectral-sum-rule diagnostics.
- Demonstrated 2,000,000× compression on a 10D benchmark (1.1 trillion grid points → 530K TT parameters). Showcase: github.com/XiaoJiang-Phy/PolaronTCISolver-Showcase.

`CrystalCanvas` 07/2025–present

An open-source GPU-accelerated scientific visualization application for crystal structures, volumetric fields, phonons, Wannier models, and reciprocal space — built for publication-quality figures. Repository: GitHub repository.

Doctoral Research Contributions

- **Crystal Structure Modulation of Polaron Properties in $BiVO_4$** → *J. Phys. Chem. C* 129, 19190 (2025)
  - Systematically investigated electron and hole polarons in three main crystal phases of $BiVO_4$ using *ab initio* calculations.
  - Identified Holstein-type electron and Fröhlich-type hole polarons and showed that variations in LO-phonon scattering strength lead to differing hole mobilities, traced to phase-dependent dielectric tensors and Born effective charges.
- **Electron–Hole Asymmetry and Polaron Transport in Defect-Free CeO_2** → *Phys. Chem. Chem. Phys.* 27, 21673 (2025)
  - Revealed a striking electron–hole asymmetry in defect-free CeO_2: electrons form Holstein polarons while holes behave as Fröhlich polarons, driven by intrinsic electron–phonon coupling rather than defect-induced mechanisms.
  - Showed that Fröhlich-polaron transport of holes is governed by both acoustic deformation potentials and high-frequency LO phonons.
- **Multi-Scale Computational Framework for Photoelectrode Design** → *Appl. Phys. Lett.* 120, 063901 (2022)
  - Designed and implemented a unified computational framework connecting first-principles calculations with macroscopic device models.
  - Applied the framework to predict optimal film thickness and guide the rational design of metal-oxide photoelectrodes, bridging microscopic theory with macroscopic performance.

Selected Collaborative Research

- **Catalytic Effects of Gray Gallium Nitride** (*ACS Nano*, 2025). Provided theoretical guidance on the simulation of interfacial charge transfer dynamics, including calculations of adsorption energies, Fermi level shifts, and work function changes.
- **Ferroelectric-Modulated Photoelectrodes** (*Adv. Funct. Mater.*, 2024). Developed optoelectronic modelling code and performed simulations of photocarrier distributions to inform structural device design.
- **Gas Adsorption Mechanisms on WO_3 Surfaces** (*J. Phys. Chem. Lett.*, 2020). Provided guidance on first-principles DFT calculation setup and assisted in analysing surface reconstruction, electron localisation, and charge transfer mechanisms.

Teaching, Mentoring and Recognition

- **Teaching Assistant**, *General Physics* (Undergraduate Course), Lanzhou University.
- **Undergraduate Student Mentor**, advised students on final-year thesis projects and scientific innovation projects.
- **Poster presentation** at the Chinese Physical Society (CPS) Fall Meeting (2021).
- **University-level Graduate Scholarship**, Lanzhou University (Multiple Awards).

Technical Expertise

Programming	Python, C++ (C++23), Julia
Software Eng. & HPC	CMake, Git/GitHub Workflows, CI/CD, MPI/OpenMP, Linux HPC
Scientific Software	Quantum ESPRESSO, VASP, Wannier90, EPW, Yambo, Elk, TRIQS (Contributor to `dft_tools`), `PolaronTCISolver` (Core Developer), `CrystalCanvas` (Core Developer)
Theoretical Models	DFT, DFPT, DMFT, GW/BSE, Tensor Cross Interpolation (TCI) / Tensor Train, Feynman Diagrammatic Methods, Polaron Theory, Boltzmann Transport
Languages	Chinese (Native); English (Proficient)

Publications

*(*Corresponding author)*

First-Author Publications

1. **Jiang, X.**, Cheng, X., Liu, Z., Ding, L., Han, W.* "First-Principles Study of Polarons in Multiple Crystal Phases of Bismuth Vanadate." *The Journal of Physical Chemistry C* 129.42, 19190-19198 (2025).
2. **Jiang, X.**, Cheng, X., Liu, Z., Ding, L., Han, W.* "Understanding Polaron Dynamics in CeO_2 for Advanced Catalytic Material Design." *Physical Chemistry Chemical Physics* 27, 21673-21682 (2025).
3. **Jiang, X.**, Cheng, X., Zhang, Z.*, et al. "Computation-assisted performance optimization for photoelectrochemical photoelectrodes." *Applied Physics Letters* 120, 063901 (2022).
4. **Jiang, X.**, Zhang, Z., Mei, J., et al. "Carbon quantum dots based charge bridge between photoanode and electrocatalysts for efficiency water oxidation performance." *Electrochimica Acta* 273, 208-215 (2018).

Co-Authored Publications

5. Liu, Z., **Jiang, X.**, Cheng, X., et al. "Catalytic Effect of Gray Gallium Nitride for Hydrogen Peroxide Generation: Insights into Mechanical Stimuli-Driven Semiconductor–Liquid Interface Alteration." *ACS Nano* 19.37, 33361–33371 (2025).
6. Shao, J., Liu, H., **Jiang, X.**, et al. "High Performance Ferroelectric-modulated Photoelectrodes by Using Grid Charge Collector." *Advanced Functional Materials* 34.32, 2316409 (2024).
7. Sun, M., Chen, W., **Jiang, X.**, et al. "Optoelectrical Regulation of $CuBi_2O_4$ Photocathode via Photonic Crystal Structure for Solar-Fuel Conversion." *ACS Applied Materials & Interfaces* 14.38, 43946-43954 (2022).
8. Liu, B., **Jiang, X.**, Jiang, X., et al. "Z-Scheme Photocarrier Transfer Realized in Tungsten Oxide-Based Photocatalysts by Combining with Bismuth Vanadate Quantum Dots." *Inorganic Chemistry* 60.5, 3057-3064 (2021).
9. Ma, Y., **Jiang, X.**, Sun, R., et al. "Z-scheme $Bi_2O_{2.33}/Bi_2S_3$ heterojunction nanostructures for photocatalytic overall water splitting." *Chemical Engineering Journal* 382, 123020 (2020).
10. Cheng, X., **Jiang, X.**, Tao, K., et al. "Microscopic Nature of Gas Adsorption on WO_3 Surfaces: Electron Interaction and Localization." *The Journal of Physical Chemistry Letters* 11.21, 9070-9078 (2020).
11. Zhang, Z., **Jiang, X.**, Mei, J., et al. "Improved photoelectrocatalytic hydrogen generation through $BiVO_4$ quantum-dots loaded on nano-structured SnO_2 and modified with carbon quantum-dots." *Chemical Engineering Journal* 331, 48-53 (2018).